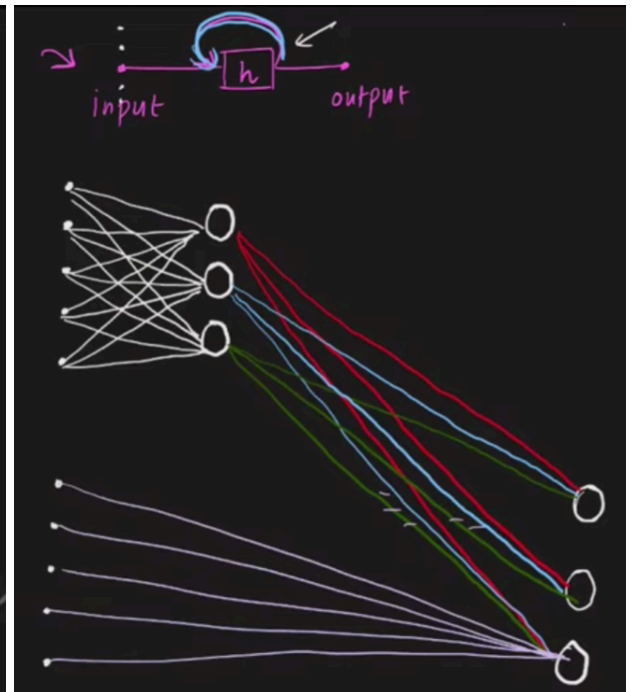
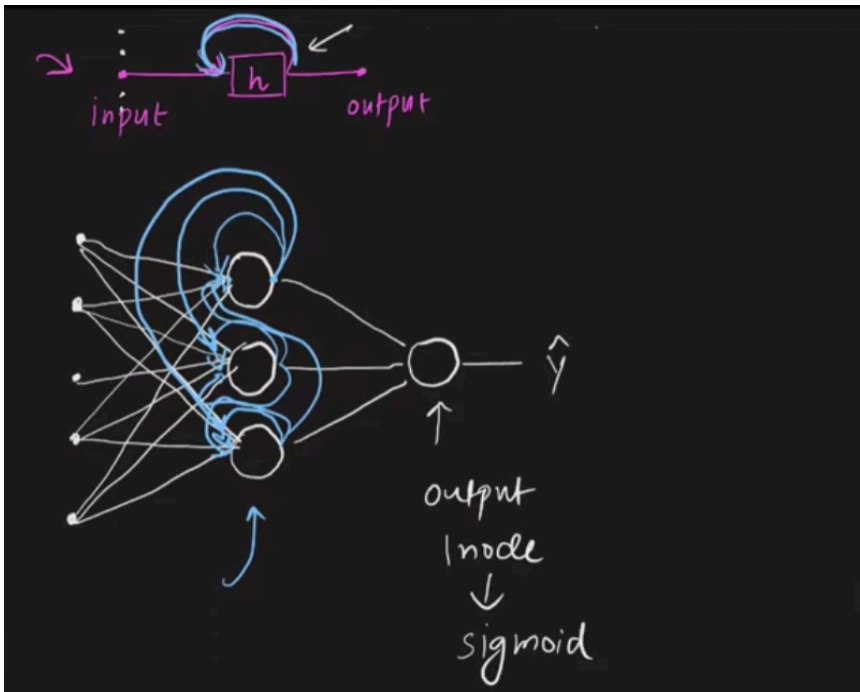


## 1. RNN internal structure



## 2. Cross-entropy Intuition

### The exact mapping

For one data point, cross-entropy loss is

$$\mathcal{L} = -\log(p)$$

where

$p = \hat{y}_{\text{true}}$  = predicted probability of the **correct class**.

So the penalty is a **logarithmic function** of the confidence.

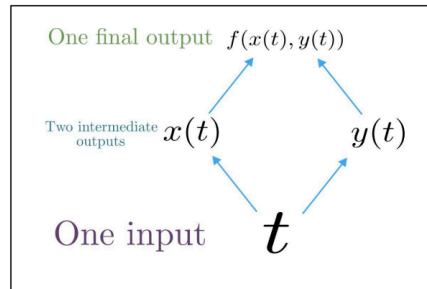
### Probability → Penalty table

| Predicted prob. $p$ | Loss = $-\log(p)$    | Intuition                    |
|---------------------|----------------------|------------------------------|
| 1.0                 | 0.000                | Perfect, no penalty          |
| 0.9                 | 0.105                | Very confident, tiny penalty |
| 0.7                 | 0.357                | Reasonable confidence        |
| 0.5                 | 0.693                | "Coin flip" uncertainty      |
| 0.1                 | 2.303                | Very wrong                   |
| 0.01                | 4.605                | Extremely wrong              |
| $\rightarrow 0$     | $\rightarrow \infty$ | Impossible prediction        |

👉 Penalty grows fast as confidence drops.

### 3. Multivariable-chain rule

## Multivariable Chain Rule



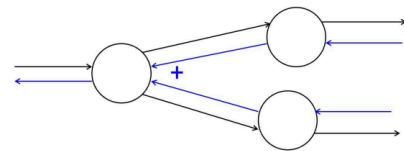
Source:  
<https://www.khanacademy.org/math/multivariable-calculus/multivariable-derivatives/differentiating-vector-valued-functions/a/multivariable-chain-rule-simple-version>

- Given a multivariable function  $f(x, y)$ , and two single variable functions  $x(t)$  and  $y(t)$ , here's what the multivariable chain rule says:

$$\frac{d}{dt} f(x(t), y(t)) = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt}$$

Derivative of composition function

### Gradients sum at outward branches



$$a = x + y$$

$$b = \max(y, z)$$

$$f = ab$$

$$\frac{\partial f}{\partial y} = \frac{\partial f}{\partial a} \frac{\partial a}{\partial y} + \frac{\partial f}{\partial b} \frac{\partial b}{\partial y}$$

### 4. Jacobian Derivation

#### Deriving the Jacobian

The RNN hidden state update is:

$$h^{(j)} = f(W_{hh} h^{(j-1)} + W_{hx} x^{(j)})$$

Let  $z^{(j)} = W_{hh} h^{(j-1)} + W_{hx} x^{(j)}$  (the pre-activation). Then by chain rule:

$$\frac{\partial h^{(j)}}{\partial h^{(j-1)}} = \frac{\partial h^{(j)}}{\partial z^{(j)}} \cdot \frac{\partial z^{(j)}}{\partial h^{(j-1)}}$$

**First term** — since  $f$  is applied elementwise, the derivative of  $h^{(j)}$  w.r.t.  $z^{(j)}$  is diagonal:

$$\frac{\partial h^{(j)}}{\partial z^{(j)}} = \text{diag}[f'(z^{(j)})]$$

**Second term** —  $z^{(j)} = W_{hh} h^{(j-1)} + \dots$  is linear in  $h^{(j-1)}$ , so:

$$\frac{\partial z^{(j)}}{\partial h^{(j-1)}} = W_{hh}$$

Its transpose appears because we're computing the Jacobian (mapping gradient vectors), giving  $W_{hh}^T$ .

Combining:

$$\frac{\partial h^{(j)}}{\partial h^{(j-1)}} = W_{hh}^T \cdot \text{diag}[f'(W_{hh} h^{(j-1)} + W_{hx} x^{(j)})]$$

### Why the first term is a diagonal matrix

The key is that  $f$  is applied **elementwise**. That means each output component  $h_i^{(j)}$  depends **only** on  $z_i^{(j)}$ , not on any other  $z_k^{(j)}$ .

So when you take the partial derivative:

$$\left[ \frac{\partial h^{(j)}}{\partial z^{(j)}} \right]_{ik} = \frac{\partial h_i^{(j)}}{\partial z_k^{(j)}}$$

- When  $i = k$ : this equals  $f'(z_i^{(j)})$  ✓
- When  $i \neq k$ : this equals 0, because  $h_i$  doesn't depend on  $z_k$  ✓

So the full Jacobian matrix looks like:

$$\frac{\partial h^{(j)}}{\partial z^{(j)}} = \begin{bmatrix} f'(z_1) & 0 & \cdots & 0 \\ 0 & f'(z_2) & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & f'(z_n) \end{bmatrix}$$

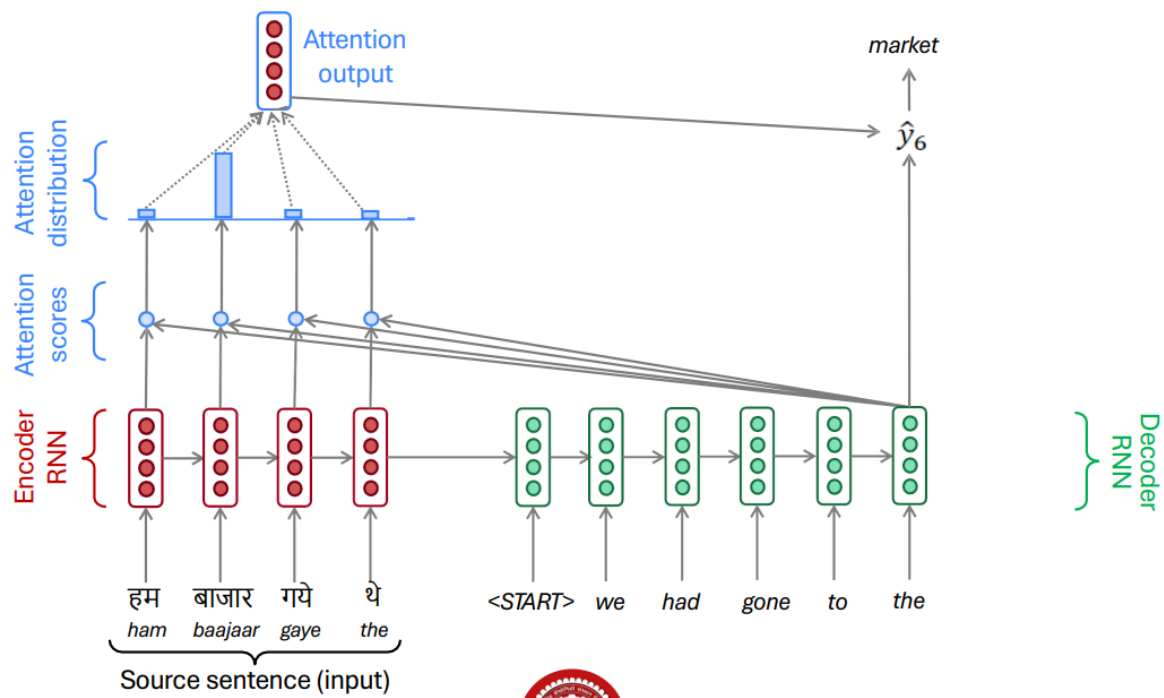
Which is exactly what  $\text{diag}[f'(z^{(j)})]$  means — a matrix with the derivatives on the diagonal and zeros everywhere else.

## 5. Vanishing / Exploding Gradient

| Your matrix |                |                | Raised to the 5th power |                |                | Raised to the 10th power |                |                | Raised to the 15th power |                   |                   | Raised to the 20th power |                      |                     |
|-------------|----------------|----------------|-------------------------|----------------|----------------|--------------------------|----------------|----------------|--------------------------|-------------------|-------------------|--------------------------|----------------------|---------------------|
|             | A <sub>1</sub> | A <sub>2</sub> |                         | A <sub>1</sub> | A <sub>2</sub> |                          | A <sub>1</sub> | A <sub>2</sub> |                          | A <sub>1</sub>    | A <sub>2</sub>    |                          | A <sub>1</sub>       | A <sub>2</sub>      |
| 1           | 0.1            | 0.5            | 1                       | 0.13056        | 0.1408         | 1                        | 0.0448004096   | 0.044695552    | 1                        | 0.014659528687616 | 0.01466060242944  | 1                        | 0.004803846016346358 | 0.00480383502123008 |
| 2           | 0.7            | 0.3            | 2                       | 0.19712        | 0.18688        | 2                        | 0.0625737728   | 0.0626786304   | 2                        | 0.020524843401216 | 0.020523769659392 | 2                        | 0.006725369029722112 | 0.00672538002483839 |

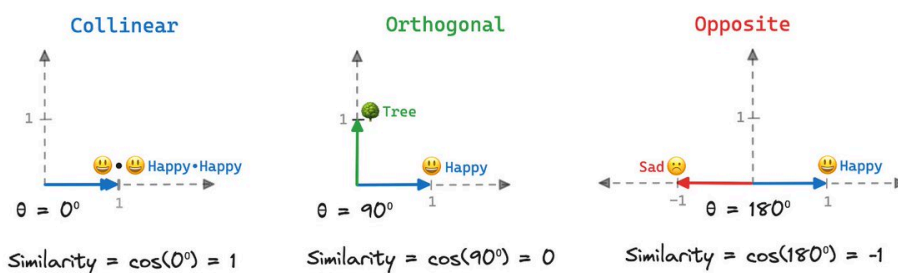
| Your matrix                                                                                                                                                         | Raised to the 5th power |                   | Raised to the 10th power |   | Raised to the 15th power |     | Raised to the 20th power |     |     |                                                                                                                                                                                        |  |                |                |   |          |         |   |          |          |                                                                                                                                                                                                                        |  |                |                |   |                  |                 |   |                  |                  |                                                                                                                                                                                                                               |  |                |                |   |                    |                   |   |                    |                   |                                                                                                                                                                                                                             |  |                |                |   |                   |                   |   |                   |                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|-------------------|--------------------------|---|--------------------------|-----|--------------------------|-----|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|----------------|----------------|---|----------|---------|---|----------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|----------------|----------------|---|------------------|-----------------|---|------------------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|----------------|----------------|---|--------------------|-------------------|---|--------------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|----------------|----------------|---|-------------------|-------------------|---|-------------------|-------------------|
| <table><tr><th></th><th>A<sub>1</sub></th><th>A<sub>2</sub></th></tr><tr><td>1</td><td>1.1</td><td>1.5</td></tr><tr><td>2</td><td>1.7</td><td>1.3</td></tr></table> |                         | A <sub>1</sub>    | A <sub>2</sub>           | 1 | 1.1                      | 1.5 | 2                        | 1.7 | 1.3 | <table><tr><th></th><th>A<sub>1</sub></th><th>A<sub>2</sub></th></tr><tr><td>1</td><td>80.66816</td><td>80.6784</td></tr><tr><td>2</td><td>91.43552</td><td>91.42528</td></tr></table> |  | A <sub>1</sub> | A <sub>2</sub> | 1 | 80.66816 | 80.6784 | 2 | 91.43552 | 91.42528 | <table><tr><th></th><th>A<sub>1</sub></th><th>A<sub>2</sub></th></tr><tr><td>1</td><td>13884.2234945536</td><td>13884.223389696</td></tr><tr><td>2</td><td>15735.4531749888</td><td>15735.4532798464</td></tr></table> |  | A <sub>1</sub> | A <sub>2</sub> | 1 | 13884.2234945536 | 13884.223389696 | 2 | 15735.4531749888 | 15735.4532798464 | <table><tr><th></th><th>A<sub>1</sub></th><th>A<sub>2</sub></th></tr><tr><td>1</td><td>2389525.9477674253</td><td>2389525.947768499</td></tr><tr><td>2</td><td>2708129.4074709658</td><td>2708129.407469892</td></tr></table> |  | A <sub>1</sub> | A <sub>2</sub> | 1 | 2389525.9477674253 | 2389525.947768499 | 2 | 2708129.4074709658 | 2708129.407469892 | <table><tr><th></th><th>A<sub>1</sub></th><th>A<sub>2</sub></th></tr><tr><td>1</td><td>411246209.0663599</td><td>411246209.0663599</td></tr><tr><td>2</td><td>466079036.9418745</td><td>466079036.9418745</td></tr></table> |  | A <sub>1</sub> | A <sub>2</sub> | 1 | 411246209.0663599 | 411246209.0663599 | 2 | 466079036.9418745 | 466079036.9418745 |
|                                                                                                                                                                     | A <sub>1</sub>          | A <sub>2</sub>    |                          |   |                          |     |                          |     |     |                                                                                                                                                                                        |  |                |                |   |          |         |   |          |          |                                                                                                                                                                                                                        |  |                |                |   |                  |                 |   |                  |                  |                                                                                                                                                                                                                               |  |                |                |   |                    |                   |   |                    |                   |                                                                                                                                                                                                                             |  |                |                |   |                   |                   |   |                   |                   |
| 1                                                                                                                                                                   | 1.1                     | 1.5               |                          |   |                          |     |                          |     |     |                                                                                                                                                                                        |  |                |                |   |          |         |   |          |          |                                                                                                                                                                                                                        |  |                |                |   |                  |                 |   |                  |                  |                                                                                                                                                                                                                               |  |                |                |   |                    |                   |   |                    |                   |                                                                                                                                                                                                                             |  |                |                |   |                   |                   |   |                   |                   |
| 2                                                                                                                                                                   | 1.7                     | 1.3               |                          |   |                          |     |                          |     |     |                                                                                                                                                                                        |  |                |                |   |          |         |   |          |          |                                                                                                                                                                                                                        |  |                |                |   |                  |                 |   |                  |                  |                                                                                                                                                                                                                               |  |                |                |   |                    |                   |   |                    |                   |                                                                                                                                                                                                                             |  |                |                |   |                   |                   |   |                   |                   |
|                                                                                                                                                                     | A <sub>1</sub>          | A <sub>2</sub>    |                          |   |                          |     |                          |     |     |                                                                                                                                                                                        |  |                |                |   |          |         |   |          |          |                                                                                                                                                                                                                        |  |                |                |   |                  |                 |   |                  |                  |                                                                                                                                                                                                                               |  |                |                |   |                    |                   |   |                    |                   |                                                                                                                                                                                                                             |  |                |                |   |                   |                   |   |                   |                   |
| 1                                                                                                                                                                   | 80.66816                | 80.6784           |                          |   |                          |     |                          |     |     |                                                                                                                                                                                        |  |                |                |   |          |         |   |          |          |                                                                                                                                                                                                                        |  |                |                |   |                  |                 |   |                  |                  |                                                                                                                                                                                                                               |  |                |                |   |                    |                   |   |                    |                   |                                                                                                                                                                                                                             |  |                |                |   |                   |                   |   |                   |                   |
| 2                                                                                                                                                                   | 91.43552                | 91.42528          |                          |   |                          |     |                          |     |     |                                                                                                                                                                                        |  |                |                |   |          |         |   |          |          |                                                                                                                                                                                                                        |  |                |                |   |                  |                 |   |                  |                  |                                                                                                                                                                                                                               |  |                |                |   |                    |                   |   |                    |                   |                                                                                                                                                                                                                             |  |                |                |   |                   |                   |   |                   |                   |
|                                                                                                                                                                     | A <sub>1</sub>          | A <sub>2</sub>    |                          |   |                          |     |                          |     |     |                                                                                                                                                                                        |  |                |                |   |          |         |   |          |          |                                                                                                                                                                                                                        |  |                |                |   |                  |                 |   |                  |                  |                                                                                                                                                                                                                               |  |                |                |   |                    |                   |   |                    |                   |                                                                                                                                                                                                                             |  |                |                |   |                   |                   |   |                   |                   |
| 1                                                                                                                                                                   | 13884.2234945536        | 13884.223389696   |                          |   |                          |     |                          |     |     |                                                                                                                                                                                        |  |                |                |   |          |         |   |          |          |                                                                                                                                                                                                                        |  |                |                |   |                  |                 |   |                  |                  |                                                                                                                                                                                                                               |  |                |                |   |                    |                   |   |                    |                   |                                                                                                                                                                                                                             |  |                |                |   |                   |                   |   |                   |                   |
| 2                                                                                                                                                                   | 15735.4531749888        | 15735.4532798464  |                          |   |                          |     |                          |     |     |                                                                                                                                                                                        |  |                |                |   |          |         |   |          |          |                                                                                                                                                                                                                        |  |                |                |   |                  |                 |   |                  |                  |                                                                                                                                                                                                                               |  |                |                |   |                    |                   |   |                    |                   |                                                                                                                                                                                                                             |  |                |                |   |                   |                   |   |                   |                   |
|                                                                                                                                                                     | A <sub>1</sub>          | A <sub>2</sub>    |                          |   |                          |     |                          |     |     |                                                                                                                                                                                        |  |                |                |   |          |         |   |          |          |                                                                                                                                                                                                                        |  |                |                |   |                  |                 |   |                  |                  |                                                                                                                                                                                                                               |  |                |                |   |                    |                   |   |                    |                   |                                                                                                                                                                                                                             |  |                |                |   |                   |                   |   |                   |                   |
| 1                                                                                                                                                                   | 2389525.9477674253      | 2389525.947768499 |                          |   |                          |     |                          |     |     |                                                                                                                                                                                        |  |                |                |   |          |         |   |          |          |                                                                                                                                                                                                                        |  |                |                |   |                  |                 |   |                  |                  |                                                                                                                                                                                                                               |  |                |                |   |                    |                   |   |                    |                   |                                                                                                                                                                                                                             |  |                |                |   |                   |                   |   |                   |                   |
| 2                                                                                                                                                                   | 2708129.4074709658      | 2708129.407469892 |                          |   |                          |     |                          |     |     |                                                                                                                                                                                        |  |                |                |   |          |         |   |          |          |                                                                                                                                                                                                                        |  |                |                |   |                  |                 |   |                  |                  |                                                                                                                                                                                                                               |  |                |                |   |                    |                   |   |                    |                   |                                                                                                                                                                                                                             |  |                |                |   |                   |                   |   |                   |                   |
|                                                                                                                                                                     | A <sub>1</sub>          | A <sub>2</sub>    |                          |   |                          |     |                          |     |     |                                                                                                                                                                                        |  |                |                |   |          |         |   |          |          |                                                                                                                                                                                                                        |  |                |                |   |                  |                 |   |                  |                  |                                                                                                                                                                                                                               |  |                |                |   |                    |                   |   |                    |                   |                                                                                                                                                                                                                             |  |                |                |   |                   |                   |   |                   |                   |
| 1                                                                                                                                                                   | 411246209.0663599       | 411246209.0663599 |                          |   |                          |     |                          |     |     |                                                                                                                                                                                        |  |                |                |   |          |         |   |          |          |                                                                                                                                                                                                                        |  |                |                |   |                  |                 |   |                  |                  |                                                                                                                                                                                                                               |  |                |                |   |                    |                   |   |                    |                   |                                                                                                                                                                                                                             |  |                |                |   |                   |                   |   |                   |                   |
| 2                                                                                                                                                                   | 466079036.9418745       | 466079036.9418745 |                          |   |                          |     |                          |     |     |                                                                                                                                                                                        |  |                |                |   |          |         |   |          |          |                                                                                                                                                                                                                        |  |                |                |   |                  |                 |   |                  |                  |                                                                                                                                                                                                                               |  |                |                |   |                    |                   |   |                    |                   |                                                                                                                                                                                                                             |  |                |                |   |                   |                   |   |                   |                   |

## 6. Seq-to-Seq with Attention



## 7. Cosine Similarity Intuition

### A Geometric intuition to vector similarity! 🚀



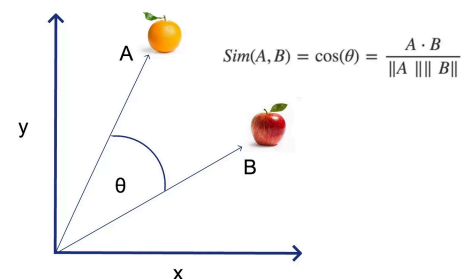
Collinear: The vectors/words are same!

Orthogonal: The vectors/words are totally un-related!

Opposite: The vectors/words are exact opposites!

@akshay\_pachaar

### Cosine Similarity



8.